

ATC Radio-over-IP Gateway

Airborne VHF + HF RoIP over Starlink / aircraft IP

Pilot headset > gateway > Starlink > ATC IP network > controller

STM32H743 · LAN8742A Ethernet · TLV320AIC23B stereo codec

Transformer-isolated audio · opto PTT / COR · 28 V aircraft bus

Project	ATC RoIP Gateway (VHF + HF, airborne)
Revision	A — schematic release
Date	10 September 2026
Prepared by	eng.ahmad@gmail.com · generated with KiCad 10
Design status	ERC clean · netlist verified · PCB routed, 0 DRC electrical
Classification	Prototype / experimental — no TSO, DO-160, DO-178C

1 Executive summary

This document specifies an airborne **Radio-over-IP (RoIP) gateway** that bridges two aircraft transceivers — a VHF COM and an HF radio — to an IP network carried by a Starlink aviation terminal. Voice is streamed as RTP with push-to-talk (PTT) and squelch/COR signaling in the style of EUROCAE ED-137, so a remote operator or ATC-side voice system can key and monitor the aircraft radios from anywhere on the network.

A market survey (Section 2) found no existing certified product with this scope: ground-side ATC VoIP is mature and standardized (ED-137), airborne VHF relays exist only in the UAV/BVLOS niche, and no airborne HF RoIP gateway was found at all. The hardware here fills that gap at prototype level: a single PCB built around an STM32H743 microcontroller, a LAN8742A Ethernet PHY, and a TLV320AIC23B stereo codec whose left and right channels carry the VHF and HF radios respectively, with full galvanic isolation toward both radios.

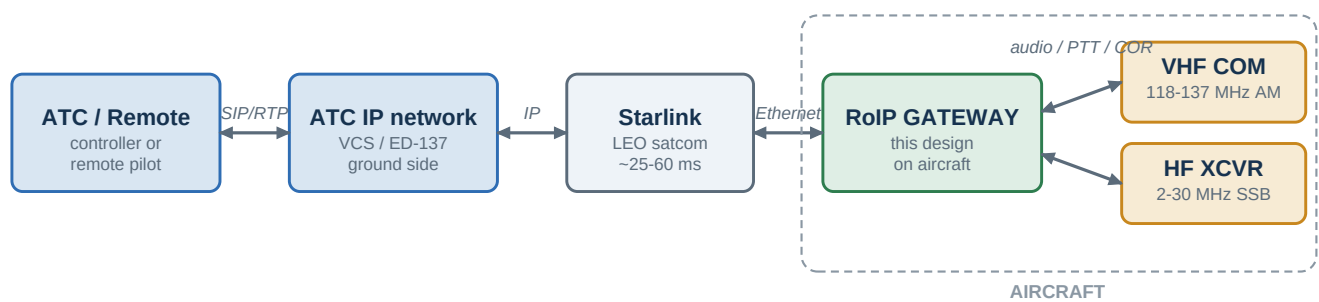
The complete schematic set was designed in **KiCad** (six hierarchical sheets, Appendix A), passes electrical rule check with zero errors and zero warnings, and its netlist has been machine-verified on fourteen critical signal paths.

2 Market context — what exists today

Segment	Status	Notes
Ground ATC RoIP	Mature, standardized	EUROCAE ED-137 (SIP + RTP with PTT/SQU header extensions); deployed under SESAR / FAA VoICE; vendors: R&S, Frequentis, Jotron.
Airborne VHF RoIP	Niche (UAV / BVLOS)	Orbit Airborne Radio Gateway, Mimer SoftRadio, SyTech: remote pilot talks to ATC through a VHF radio on the UAV via datalink / satcom.
Airborne HF RoIP	Not found	HF-over-IP exists only at ground stations of oceanic networks. A genuine product gap.
Voice over Starlink	Not certified	Starlink carries broadband / VoIP; certified long-range voice remains SATVOICE (Inmarsat / Iridium), itself still an HF backup, not a replacement.

Target applications, in order of practicality: (1) UAS / optionally piloted aircraft where the remote pilot keys the aircraft radios over Starlink; (2) flight test and special-mission audio streaming and recording; (3) experimental-aircraft backup audio paths; (4) airborne HF remote-operation trials.

3 System architecture



Stream:
RTP/UDP · 20 ms frames · Opus 16 kbit/s or G.711 μ-law (ED-137 Radio profile) · PTT + squelch in RTP header extension · adaptive 40-120 ms jitter buffer

Figure 1 — End-to-end signal chain

The gateway sits on the aircraft LAN behind the Starlink router and terminates an RTP session per radio channel. On the radio side it behaves like a remote operator position: it injects microphone audio and keys PTT, and it captures receiver audio together with the radio's carrier-operated squelch (COR) state. Latency budget over Starlink is typically 25–60 ms one-way plus jitter, which the adaptive jitter buffer absorbs; this is acceptable for remote-pilot and monitoring use but is one of the reasons tactical ATC certification is a long-term goal rather

than a rev-A claim.

4 Hardware design

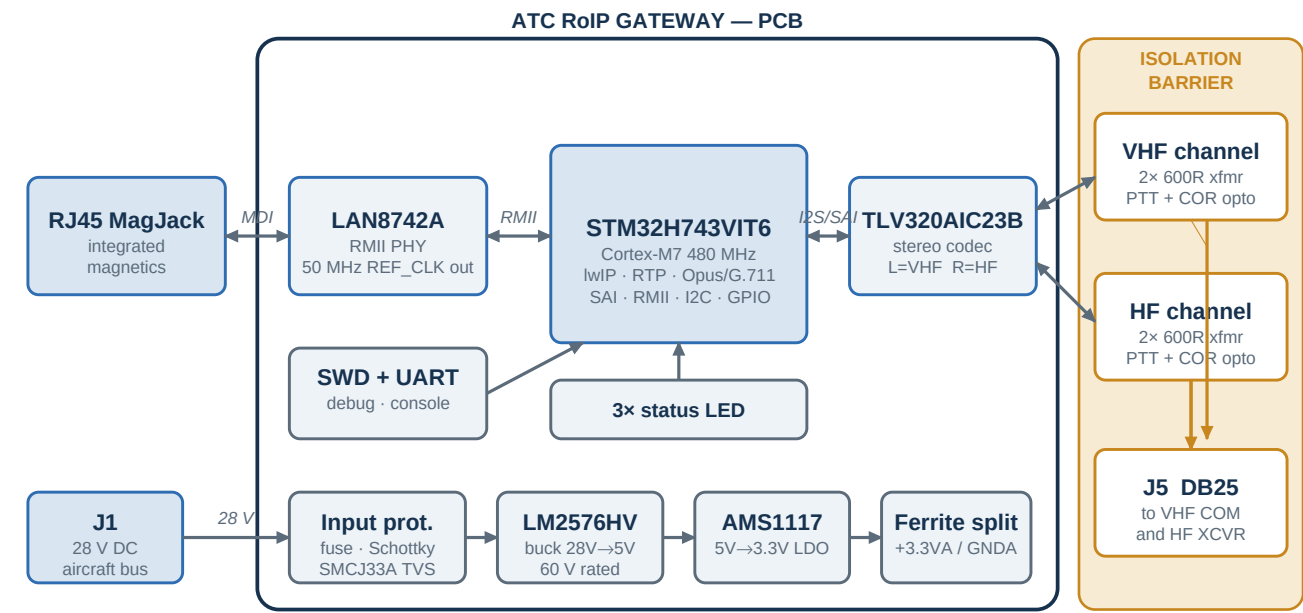


Figure 2 — Board block diagram (blocks map 1:1 to schematic sheets)

4.1 Processor and network

The STM32H743VIT6 (Cortex-M7, 480 MHz, LQFP-100) provides the Ethernet MAC, the SAI serial-audio interface, and enough compute headroom for lwIP plus two simultaneous Opus encoder/decoder pairs. The LAN8742A PHY connects over RMII; it is strapped (nINTSEL low) to source the 50 MHz RMII reference clock, saving a MCU clock output. The RJ45 jack has integrated magnetics; its center taps are biased from +3V3 through a ferrite bead per the PHY's drive scheme.

4.2 Audio path

A single TLV320AIC23B stereo codec serves both radios: **left channel = VHF**, **right channel = HF**, in both directions, giving perfectly aligned sampling from one SAI port. Airband AM and HF SSB voice occupy 300–2500 Hz, so 8 kHz sampling suffices (16 kHz recommended with Opus). The codec is controlled over I2C at address 0x1A and runs as I2S slave from SAI1 (MCLK, BCLK, FS).

4.3 Power

The 28 V DC aircraft bus enters through a fuse, a series Schottky for reverse polarity, and an SMCJ33A transient clamp. An LM2576HV-5.0 buck (60 V maximum input) survives bus transients and feeds an AMS1117-3.3 LDO. Ferrite beads split a dedicated +3.3VA / GNDA analog domain for the codec. Full DO-160 section 16/17 power quality (200 ms holdup, surge waveforms) requires an additional upstream stage and is deliberately out of scope for rev A.

4.4 Key components

Block	Part	Function
MCU	STM32H743VIT6	lwIP + RTP + codec control, RMII MAC, SAI
Ethernet PHY	LAN8742A	RMII, REF_CLK-out strap, 1.5 kΩ MDIO pull-up
RJ45	Amphenol RJMG1BD3B8K1ANR	Integrated magnetics + LEDs
Codec	TLV320AIC23BPW	Stereo ADC/DAC, I2S slave, I2C 0x1A

Block	Part	Function
Audio isolation	600:600 Ω transformers $\times 4$	RX + TX per channel (DIP-4 placeholder footprint)
PTT / COR	PC817 $\times 4$	Opto key-out and squelch sense per channel
Buck	LM2576HVS-5.0	28 V $\rightarrow$ 5 V, 3 A, 60 V rated
LDO	AMS1117-3.3	5 V $\rightarrow$ 3.3 V digital
Protection	SS54 + SMCJ33A	Reverse polarity + transient clamp

5 Radio interface (isolation barrier)

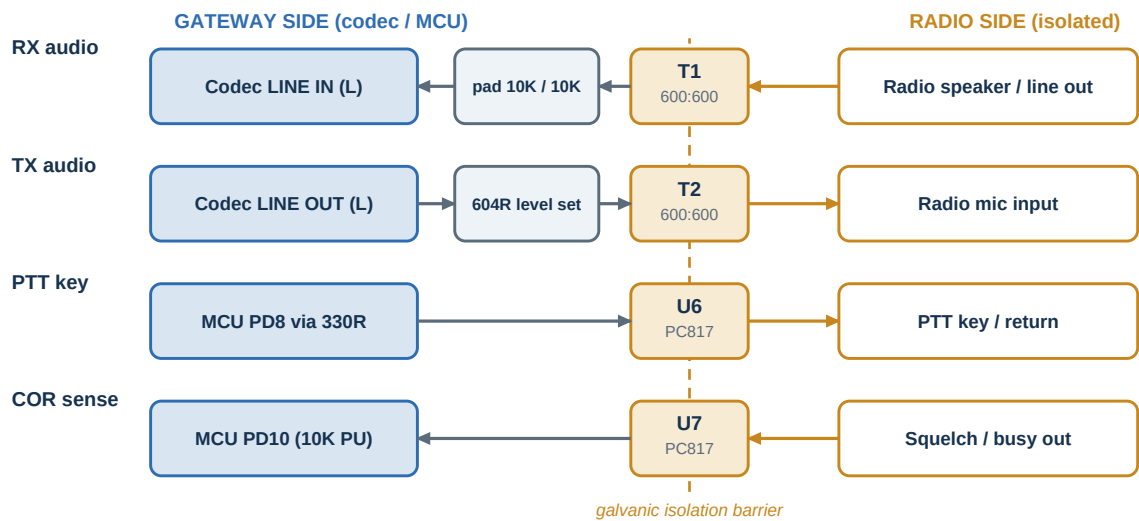


Figure 3 — One radio channel (VHF shown; HF identical)

Every line to the radios crosses a galvanic isolation barrier, killing ground loops between the gateway, the airframe, and the radio installs. Receive and transmit audio each pass through a 600:600 Ω transformer; the receive side is padded 10 k Ω /10 k Ω before the codec line input and the transmit side has a 604 Ω series level-set resistor (trim per radio's mic sensitivity). PTT is keyed by a PC817 optocoupler whose collector–emitter closes the radio's PTT line to its own ground return (≤ 50 mA / 35 V — fit a relay or MOSFET buffer for HF couplers that key more current, and a TVS across the key line for antenna-tuner kick-back). The radio's squelch/busy output drives a second opto through 4.7 k Ω , giving the firmware a true COR input; DSP-based VOX is the fallback when a radio provides no COR line.

5.1 DB25 radio connector (J5)

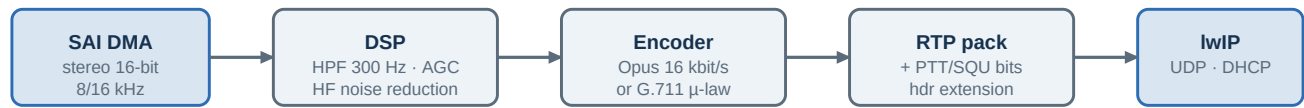
Signal	VHF pins	HF pins	Notes
RX audio hi / lo	1 / 14	6 / 19	From radio speaker or 600 Ω line out
Mic audio hi / lo	2 / 15	7 / 20	To radio mic input, level set by series R
PTT key / return	3 / 16	8 / 21	Opto collector / emitter, isolated
COR hi / lo	4 / 17	9 / 22	Radio squelch/busy output drives opto LED
Chassis / common	13, 25	13, 25	Plus connector shell

5.2 MCU pin allocation

Function	Pins
RMII to LAN8742A	PA1 REF_CLK · PA2 MDIO · PC1 MDC · PA7 CRS_DV · PC4/PC5 RXD0/1 · PB11 TX_EN · PB12/PB13 TXD0/1 · PE10 PHY reset
SAI1 to codec	PE2 MCLK · PE5 BCLK · PE4 FS (LRCIN+LRCOUT) · PE6 → DIN · PE3 ← DOUT
I2C1 codec control	PB6 SCL · PB7 SDA (4.7 k Ω pull-ups)
PTT outputs	PD8 VHF · PD9 HF (330 Ω to opto LED)
COR inputs	PD10 VHF · PD11 HF (10 k Ω pull-ups, opto pulls low)
Debug / console	SWD PA13/PA14 (J2) · USART1 PA9/PA10 (J3)
Status LEDs	PD12 · PD13 · PD14

6 Firmware architecture

Radio → IP (uplink)



IP → Radio (downlink)

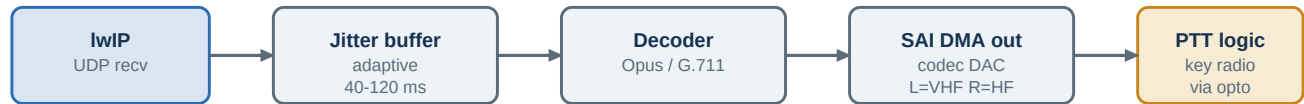


Figure 4 — Firmware dataflow (per radio channel)

- **Transport:** RTP over UDP, 20 ms frames; Opus 16 kbit/s VBR per channel, or G.711 μ -law 64 kbit/s for ED-137B Radio-profile compatibility. PTT and squelch travel in the RTP header extension (ED-137 R2S style).
- **Session:** statically configured peer for rev A; SIP session management (full ED-137 signaling) as a later increment.
- **Jitter buffer:** adaptive 40–120 ms, sized for Starlink jitter.
- **HF channel extras:** AGC, noise reduction, 300–2700 Hz bandpass, and a key-line interlock delay for antenna-coupler tune cycles.
- **VOX fallback** when no COR line is wired to a radio.
- **Config:** UART console + DHCP; small HTTP status page later.

This stack is implemented in the **firmware/** tree: ED-137-style RTP with the PTT/SQU header extension and R2S keepalive (`rtp_ed137.c`), a minimal SIP endpoint answering INVITE/ACK/BYE/OPTIONS with WG67-flavoured SDP (`sip_mini.c`), G.711 u-law (`g711.c`), the TLV320AIC23B driver (`aic23.c`), SAI DMA streaming (`sai_stream.c`), per-channel jitter buffer and PTT/COR logic (`roip_chan.c`), and HPF/AGC/VOX conditioning (`voxagc.c`). It builds with CMake + arm-none-eabi-gcc against the official STM32CubeH7 HAL/lwIP and flashes over SWD (header J2) with STM32CubeProgrammer (`flash.ps1`) or via the ROM bootloader on the J3 UART.

7 PCB implementation

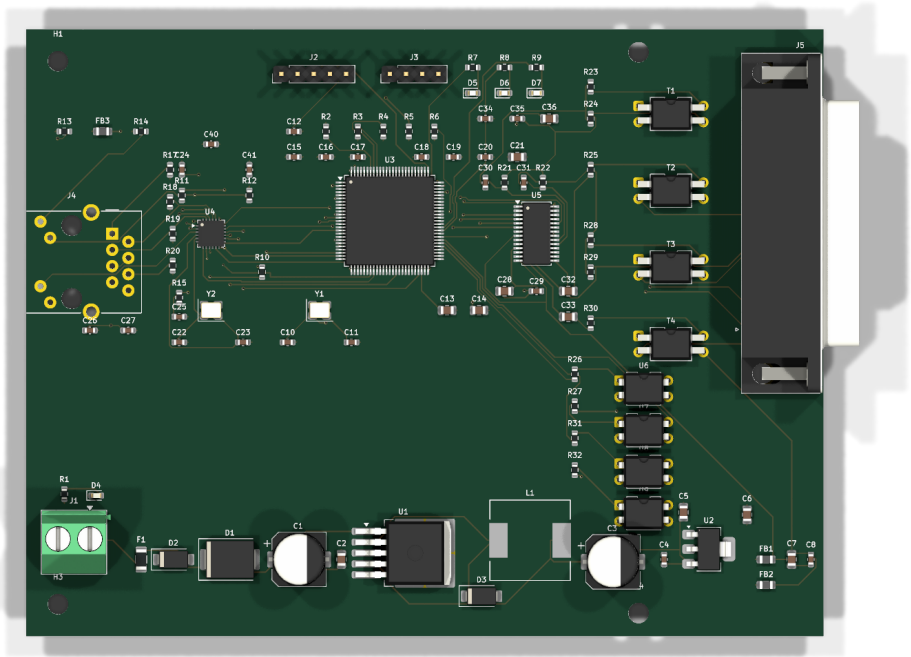


Figure 5 - Routed board, top side (125 x 95 mm, 4-layer)

The board is a 125 x 95 mm four-layer design: signals on the outer layers, solid GND (In1) and +3V3 (In2) planes inside. Floorplan: RJ45 with integrated magnetics and the LAN8742A PHY on the left edge, STM32H743 in the centre, TLV320AIC23B codec to its right, and the isolation zone - four 600 Ohm transformers, four PC817 optocouplers and the DB25 radio connector - along the right edge; the 28 V power chain runs along the bottom from the input terminal block through the buck and LDO. Routing: Freerouting autorouter constrained to the outer layers, then scripted collision-checked power fanout (86+ plane-stitching vias) and hand-verified repairs in congested areas. Rules: 0.2 mm min track, 0.15 mm clearance, 0.45/0.2 mm min via - standard 4-layer fab capability.

Manufacturing outputs (hardware/atc-roip-gateway/fab/): Gerbers, Excellon drill + map, pick-and-place CSV, and a ready-to-upload Gerber ZIP.

8 Bill of materials

Complete BOM: docs/BOM.csv - 40 line items, 100 components, with manufacturer part numbers. Summary:

References	Qty	Value	Manufacturer P/N
C1	1	100uF/63V	EEE-FK1J101P
C2	1	100nF/100V	CL21 series
C3	1	470uF/16V	EEE-FK1C471P
C4, C8, C12, C15, C16, C17,...	16	100nF	CL10 series
C5, C6, C7, C28, C32, C33, C36	7	10uF	CL21 series
C10, C11, C22, C23	4	18pF	CL10 series
C13, C14	2	2.2uF	CL21 series
C21	1	4.7uF	CL21 series

References	Qty	Value	Manufacturer P/N
C24, C30, C31	3	1uF	CL10 series
C27	1	10nF	CL10 series
D1	1	SMCJ33A	SMCJ33A
D2, D3	2	SS54	SS54
D4	1	PWR	19-217/GHC-YR1S2/3T
D5	1	STAT1	19-217/GHC-YR1S2/3T
D6	1	STAT2	19-217/GHC-YR1S2/3T
D7	1	STAT3	19-217/GHC-YR1S2/3T
F1	1	1A	0451001.MRL
FB1, FB2, FB3	3	600R@100MHz	BLM21PG600SN1D
J1	1	28VDC_BUS_IN	1715022 (MKDS 1,5/2-5,08)
J2	1	SWD	PinHeader 1x05 2.54mm
J3	1	CONSOLE_UART	PinHeader 1x04 2.54mm
J4	1	RJ45_MagJack	RJMG1BD3B8K1ANR
J5	1	TO_RADIOS	182-025-113R531 (female) or L717SDB25P
L1	1	100uH/3A	SRP1265A-101M
R1, R7, R8, R9, R13, R14, R...	8	330R	RC0603FR-07330RL
R2, R5, R6, R12, R15, R23, ...	9	10K	RC0603FR-0710KL
R3, R4, R27, R32	4	4.7K	RC0603FR-074.7KL
R10	1	12.1K	RC0603FR-0712.1KL
R11	1	1.5K	RC0603FR-071.5KL
R17, R18, R19, R20	4	49.9R	RC0603FR-0749.9RL
R21, R22	2	47K	RC0603FR-0747KL
R25, R30	2	604R	RC0603FR-07604RL
T1, T2, T3, T4	4	600:600	SP-66 (or Tamura TTC-108)
U1	1	LM2576HVS-5.0	LM2576HVS-5.0/NOPB
U2	1	AMS1117-3.3	AMS1117-3.3
U3	1	STM32H743VIT6	STM32H743VIT6
U4	1	LAN8742A	LAN8742A-CZ-TR
U5	1	TLV320AIC23BPW	TLV320AIC23BPWR
U6, U7, U8, U9	4	PC817	PC817X2NSZ9F
Y1, Y2	2	25MHz	ABM3-25.000MHZ-D2Y-T

9 Verification

Schematic, netlist, PCB and fab outputs are machine-generated and machine-checked; every artifact is reproducible from tools/.

Check	Tool	Result
Electrical rule check (all sheets)	kicad-cli sch erc	0 errors, 0 warnings
Netlist connectivity, 14 critical paths (RMII, SAI, I2C, PTT, COR, audio, power)	tools/check_netlist.py	14 / 14 pass
Power rail membership	netlist audit	+3V3 41 pins, GND 62, GNDA 17, +3.3VA 7, +5V 6
PCB connectivity	kicad-cli pcb drc	0 unconnected pads (fully routed)
PCB electrical rules (clearance, shorts, holes, edge)	kicad-cli pcb drc	0 violations
PCB cosmetics	kicad-cli pcb drc	13 silkscreen notes (connector artwork; fab-clipped)

10 Compliance and regulatory position

- **ED-137:** ED-137 is a protocol standard — compliance is earned in firmware and by conformance test. This hardware is *ED-137-ready*: G.711-capable 8 kHz codec, Ethernet, and real PTT/COR signal paths. Validation against an ED-137 test set (e.g. GL MAPS) is a firmware milestone.
- **Certification:** rev A is prototype/experimental hardware — no TSO authorization, no DO-160 qualification, no DO-178C/DO-254 artifacts. Suitable for bench, UAS, and experimental-category use.
- **Spectrum:** the gateway only keys an already-certified transceiver; transmissions on ATC frequencies require an appropriately licensed radio and operator.
- **Operational:** ATC voice over Starlink is not an approved substitute for required VHF/HF/SATVOICE equipage in controlled airspace.

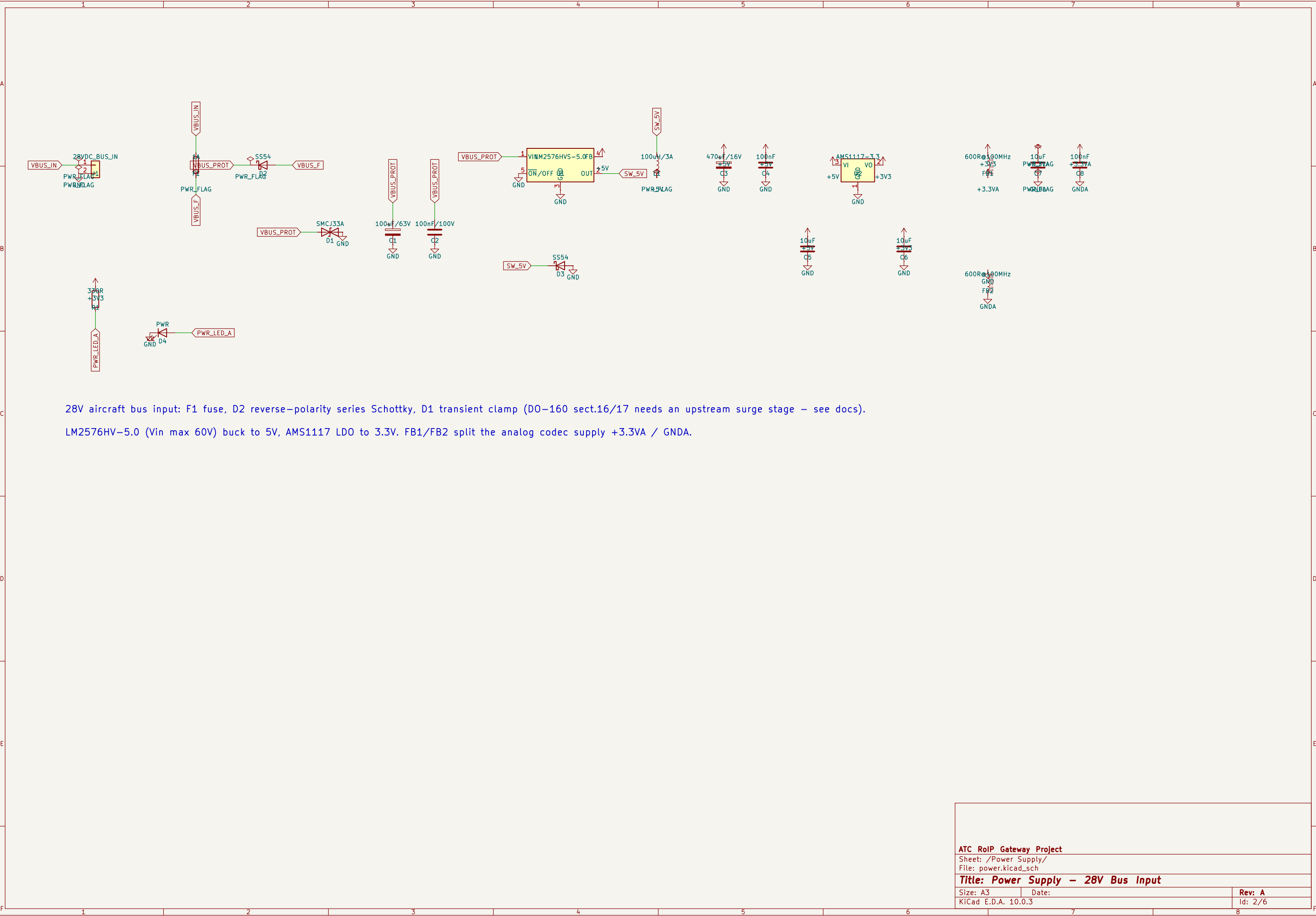
11 Next steps

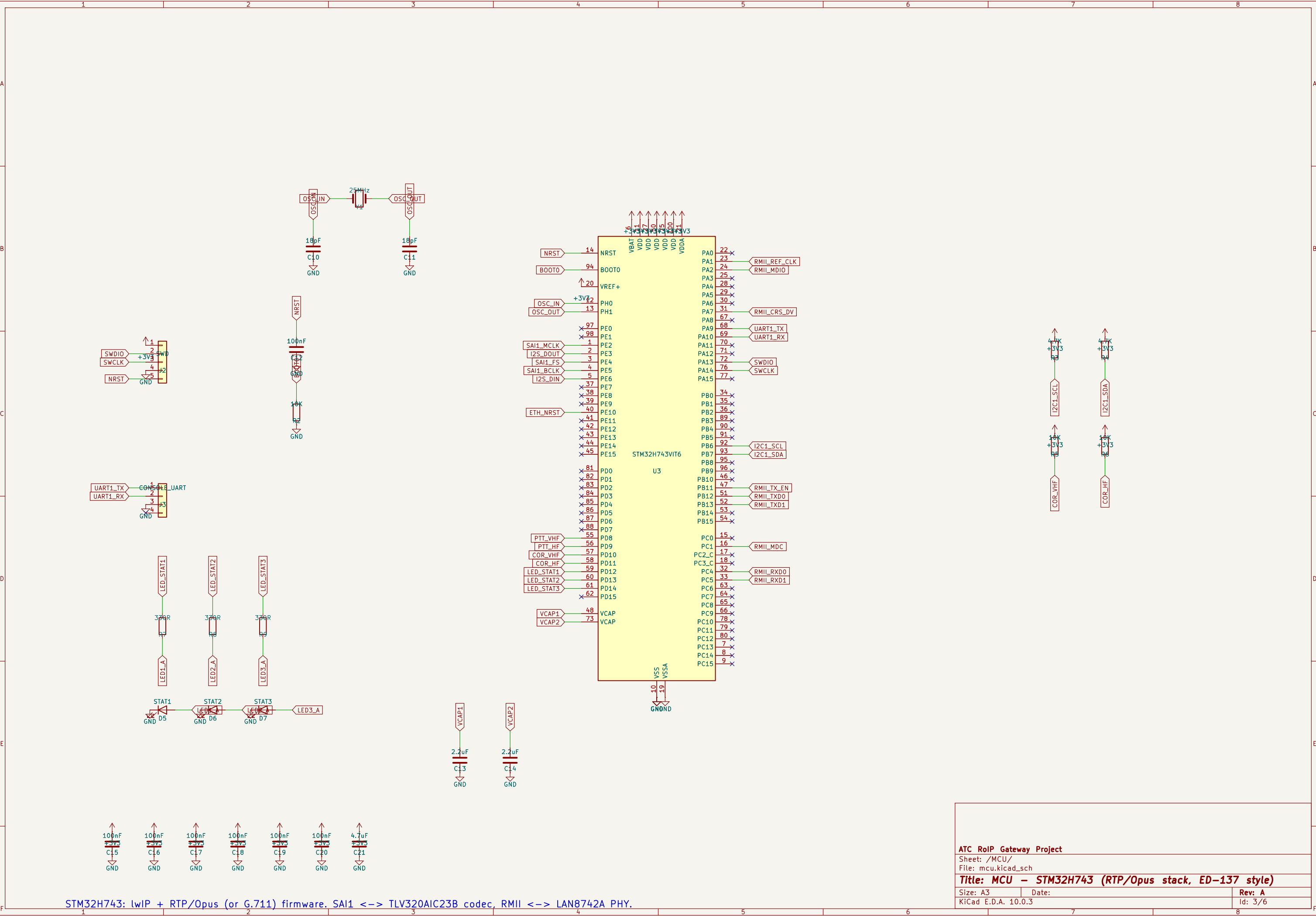
- Review radio-side levels against the specific transceivers (mic sensitivity, PTT current, HF coupler behaviour).
- Order boards from the fab ZIP; assemble, bring up power, then SWD.
- Firmware bring-up on NUCLEO-H743ZI + codec breakout before board spin.
- Bench test against an ED-137 radio emulator; then live-radio testing on the bench with dummy loads.

Appendix A — Schematic sheets

The following six A3 sheets are the complete rev A schematic as plotted from KiCad: root/index, power supply, MCU, Ethernet, audio codec, and radio interface. Every pin carries a drawn wire stub with its net label; the netlist is the authoritative artifact and is what the Section 9 checks validate.







ATC RoIP Gateway Project		
Sheet: /MCU/ File: mcu.kicad_sch		
Title: MCU - STM32H743 (RTP/Opus stack, ED-137 style)		
Size: A3	Date:	Rev: A
KiCad E.D.A. 10.0.3		Id: 3/6

STM32H743: lwIP + RTP/Opus (or G.711) firmware. SAI1 <-> TLV320AIC23B codec, RMII <-> LAN8742A PHY.

PD8/PD9 key the radio PTT optos; PD10/PD11 read COR/squelch optos (10K pull-ups R5/R6).

Id: 4/6

